Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404: Final Demo Presentation
Team 46: EchoPoint

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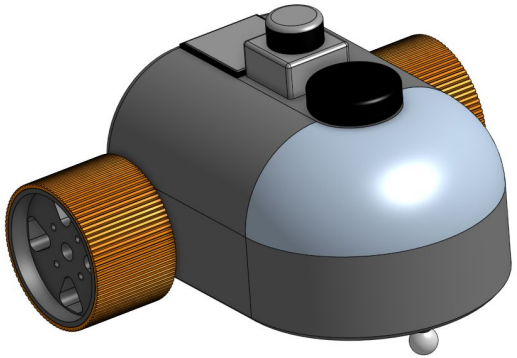
Project Overview

Problem Statement

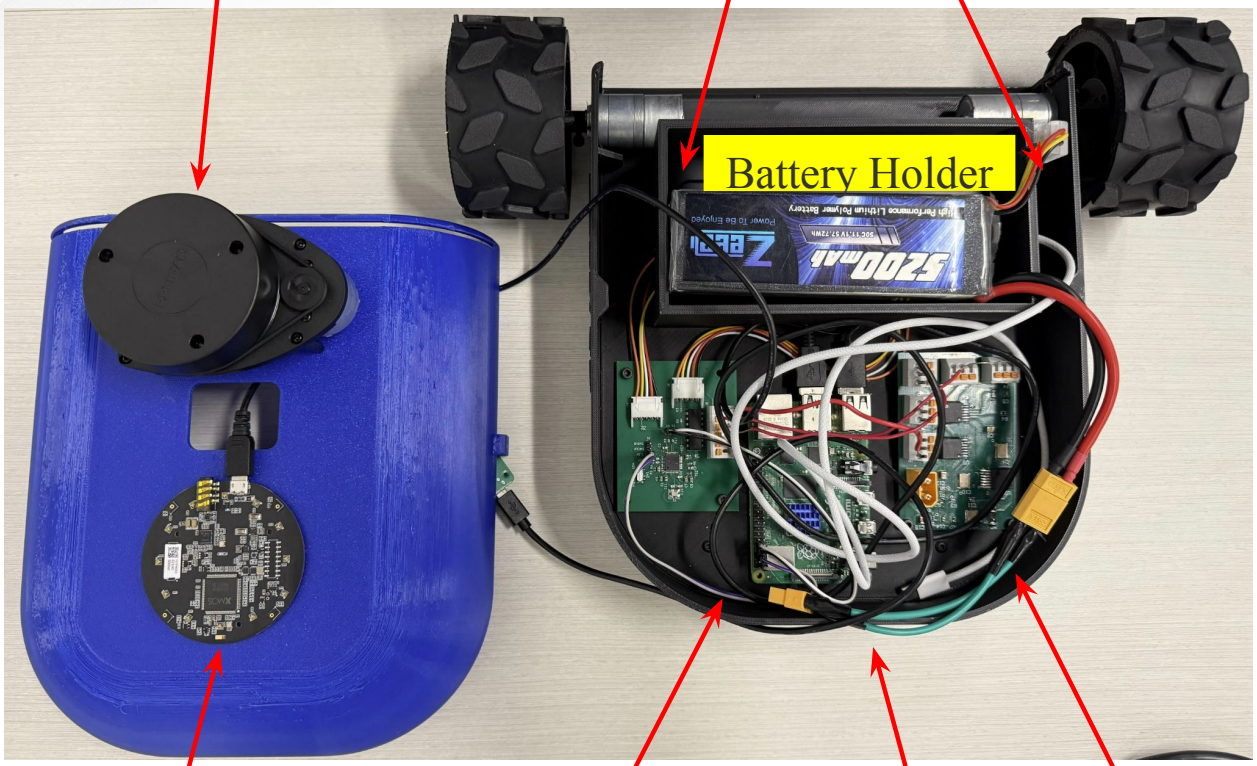
- Victims in rescue situations may be heard but not seen
- Camera-based robots fail when visibility is poor
- Need a system that locates people using sound instead of sight

Our Solution

- Mobile robot with microphone array for sound localization
- Raspberry Pi for real-time audio processing and direction detection
- STM32 microcontroller for motor control and navigation
- LiDAR for obstacle detection and safe path planning
- Robot moves toward clapping/voice source to help first responders locate individuals



Integrated Project Diagram

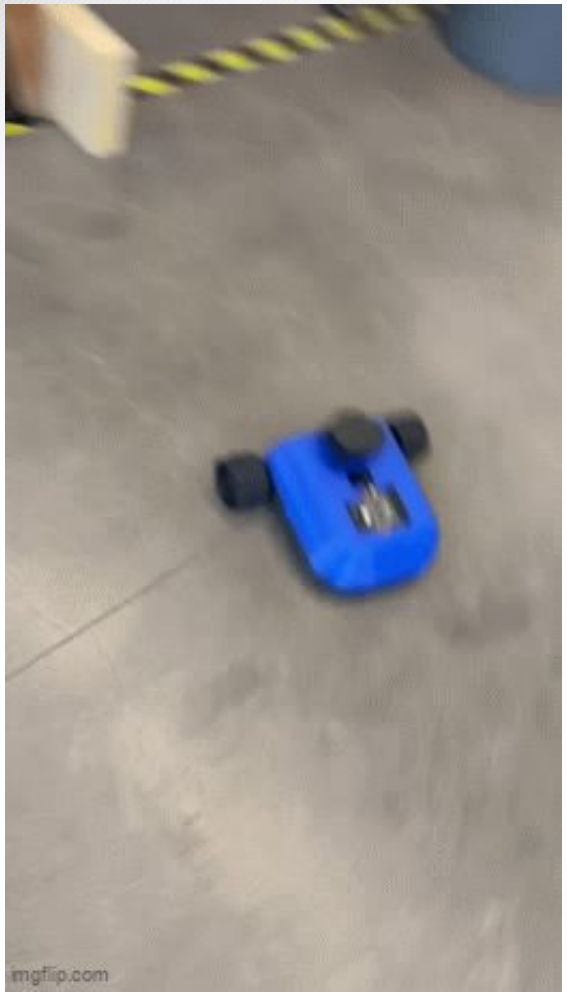


Motor Controller PCB and Chassis Accomplishments

Challenges on subsystems	Accomplishments on Echo Point
Chassis - Spacing for wires - Fell off table and cracked the chassis Motor PCB - MCU not turning on properly - MCU died and had to be replaced - Wires connecting to motors are weak - Reverse PWM on Motors - Not enough traction on wheels	- Chassis completely printed and holds everying needed in the robot - Motor PCB correctly interprets data from Pi and moves motors - Motor PCB is stable and has fail safes in case of connection loss to Pi Fail safe for motor PCB - When motors dont receive any signal for 300 ms they shut off

Motor Controller PCB and Chassis Accomplishments

Andrew Kazour



Move speed

- 3 ft/s
- Any smooth/slightly bumpy terrain
- PWM control for different speeds depending on situation

Motor reaction delay

- 15-50 ms

LiDAR Subsystem Accomplishments

Challenges on subsystems	Accomplishments on Echo Point
LiDAR Coding - Integration with Mic and Motor Code - LiDAR only had distance sensing - Realized obstacle avoidance logic had to be implemented in LiDAR code - Challenges implementing LiDAR override - Peripheral Sensing building - Left & Right turning capabilities sensitivity - Making Turning power adjustable for easier testing	- Successfully integrated microphone, LiDAR, and motor control code - Implemented functional distance sensing and obstacle avoidance - Achieved accurate distance measurements with completed validation - LiDAR system meets FSR requirements - Resolved prior issues preventing correct LiDAR operation - System now operates reliably with only minor tuning potentially required

LiDAR Subsystem Distance Sensing Accuracy

True Distance	LiDAR Sensing (avg)	Percent Error
0.25 m	0.27 m	≤1.08%
0.75 m	0.80 m	≤1.06%
1 m	1.03 m	≤1.03%
1.25 m	1.23 m	≤1.02%
4 m	3.89 m	≤1.02%
5 m	5.32 m	≤1.06%
6 m	6.02 m	≤1%
7 m	6.84 m	≤1.02%
8 m	8.03 m	≤1%

Percent Error after 60 trials

LiDAR Subsystem Avoidance Detection

Given an obstacle, whether in front, or around the peripherals, the robot will be able to sense it, and turn accordingly.

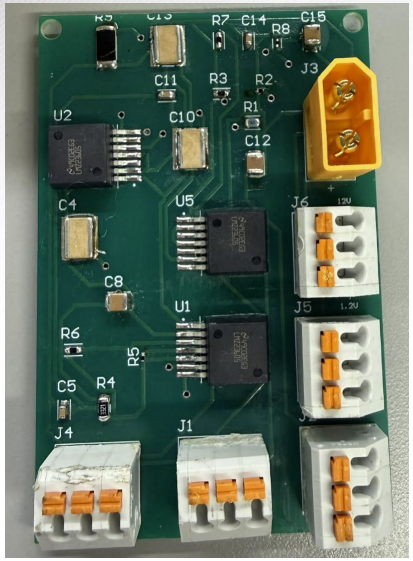
Power Subsystem Accomplishments

Arabella Aguirre

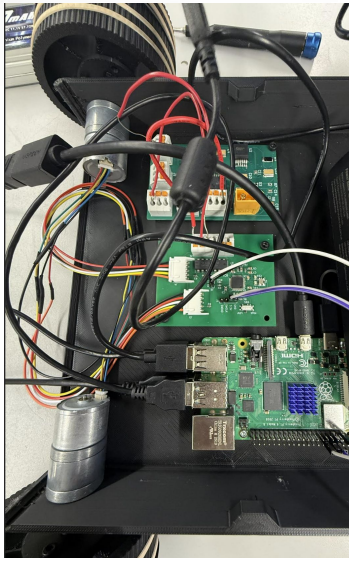
Challenges on subsystems	Accomplishments on Echo Point
- PCB did not initially output the correct voltage required for the STM32. - Voltage regulation issues affected subsystem reliability during integration - Short circuit occurred during testing - The original PCB design needed improvement to provide more stable voltage and current delivery	- Replaced faulty IC Chips to restore proper PCB functionality - Improved PCB design to provide stable voltage output of the STM32 - Increased current delivery capability for more reliable subsystem performance - Resolved previous voltage regulation issues that were preventing proper operation - Achieved stable and reliable electrical performance during testing

Power Subsystem Accomplishments Validation/Testing

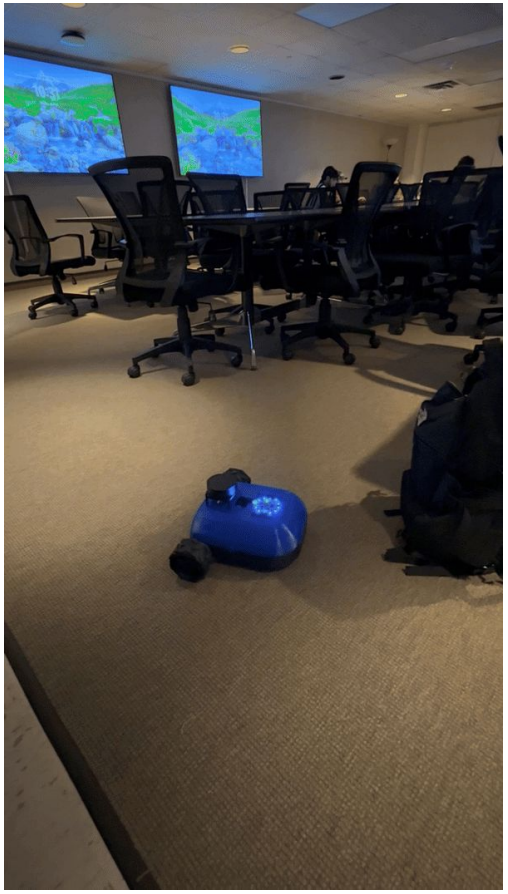
Arabella Aguirre



PCB = fully soldered

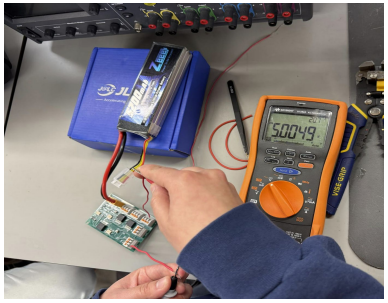


PCB = fully integrated with robot

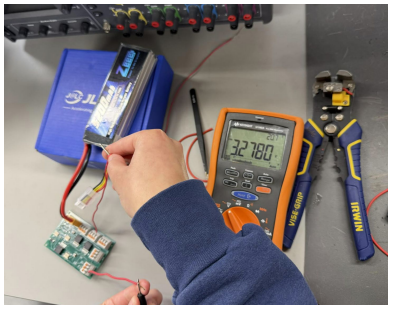


Echo Point Functioning = Power successful

12.0V



Raspberry Pi = 5.0



STM32 = 3.2V

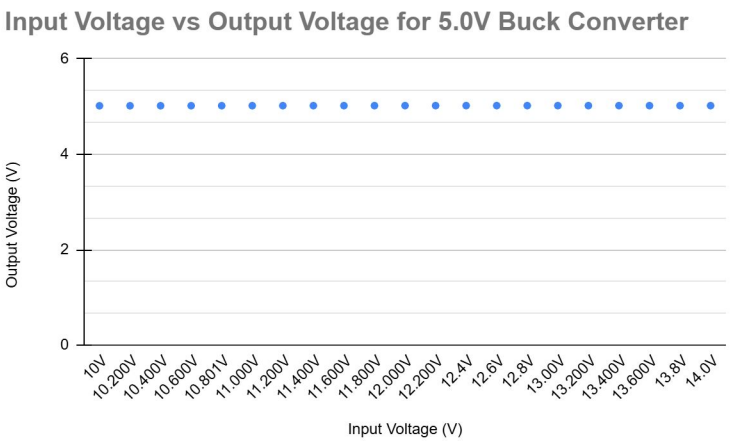
Power Subsystem Validation/Testing

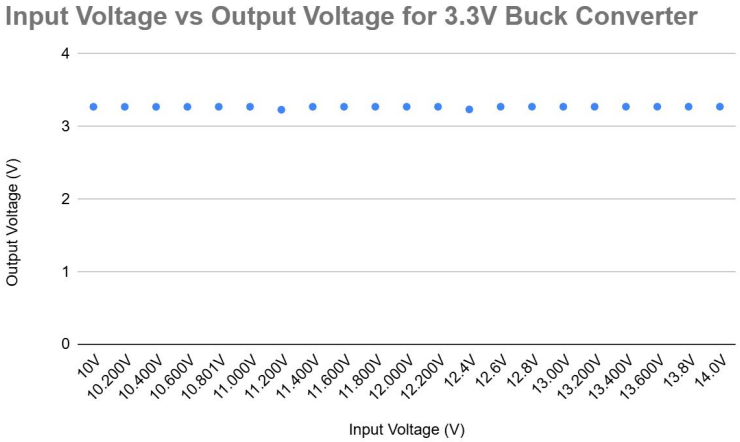
Arabella Aguirre

Open circuit/ Line testing

5V DC-DC converter	
10V	5.0172
10.200V	5.0178
10.400V	5.0181
10.600V	5.0184
10.801V	5.0188
11.000V	5.0193
11.200V	5.0195
11.400V	5.0197
11.600V	5.02
11.800V	5.0202
12.000V	5.0203
12.200V	5.0205
12.4V	5.0208
12.6V	5.0207
12.8V	5.0206
13.00V	5.0204
13.200V	5.0203
13.400V	5.0202
13.600V	5.0201
13.8V	5.02
14.0V	5.02

3.3V DC-DC converter	
10V	3.2729
10.200V	3.2722
10.400V	3.2719
10.600V	3.2724
10.801V	3.273
11.000V	3.27304
11.200V	3.2321
11.400V	3.27318
11.600V	3.2731
11.800V	3.2734
12.000V	3.273
12.200V	3.2729
12.4V	3.2365
12.6V	3.2739
12.8V	3.2738
13.00V	3.2736
13.200V	3.2736
13.400V	3.2739
13.600V	3.2741
13.8V	3.274
14.0V	3.2742





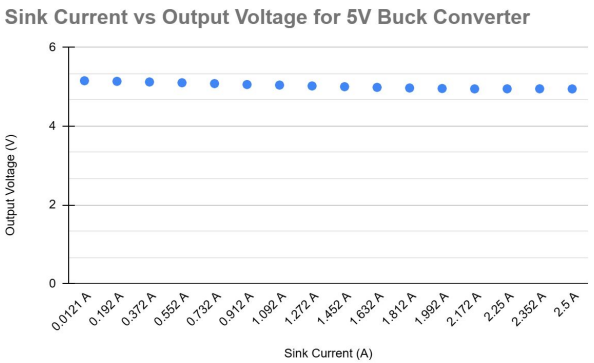
Power Subsystem Validation/Testing

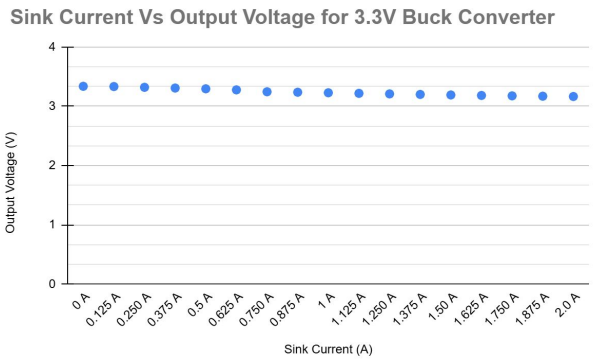
Arabella Aguirre

Electronic Load Test

Electronic Load Testing 5.1V Buck Converter	
Iout; 0.3 increments	Output Voltage
0.3 A	5.11
0.6 A	5.105
0.9 A	5.1
1.2 A	5.094
1.5 A	5.088
1.8 A	5.081
2.1 A	5.074
2.4 A	5.067
2.7 A	5.06
3.0 A	5.053
3.3 A	5.046
3.6 A	5.039
3.9 A	5.032
4.2 A	5.025
4.5 A	5.018

Electronic Load Testing 3.3V Buck Converter	
0 A	3.337
0.125 A	3.333
0.250 A	3.32
0.375 A	3.308
0.5 A	3.295
0.625 A	3.277
0.750 A	3.245
0.875 A	3.237
1 A	3.228
1.125 A	3.218
1.250 A	3.209
1.375 A	3.2
1.50 A	3.191
1.625 A	3.183
1.750 A	3.176
1.875 A	3.17
2.0 A	3.165





Microphone Subsystem Accomplishments

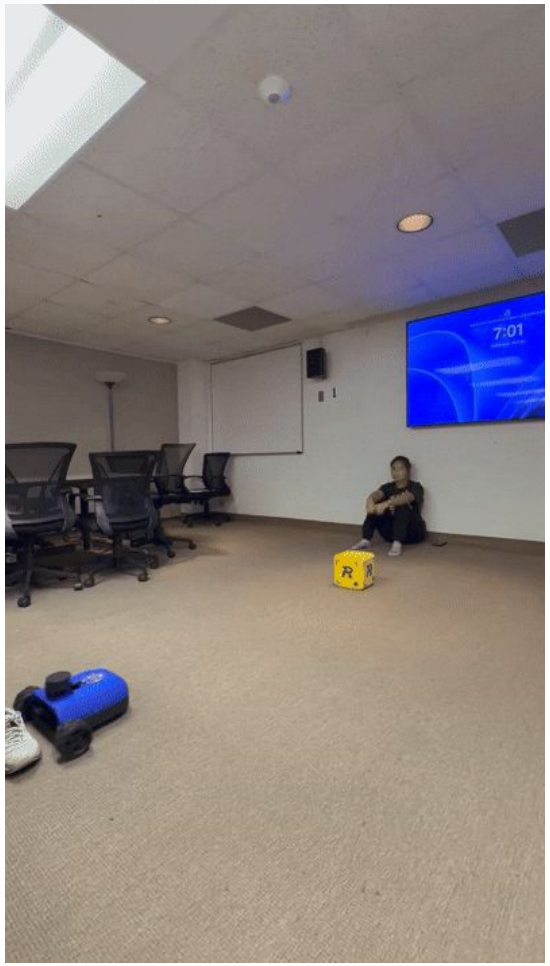
Challenges on subsystems	Accomplishments on Echo Point
- Getting the microphone subsystem to calculate the correct angle - Finding a sound that the microphone subsystem could both detect and calculate the correct direction - Rooms with lots of echo, noisy rooms, sound of motor and sled distracted the robot - When integrated, robot found it hard deciding whether to prioritize LiDAR or microphone code	- Altered microphone code to accurately calculate angle from where sound is coming from, given real time audio data - Successfully integrated microphone subsystem with LiDAR, motor controller, and power subsystems

Microphone Subsystem Validation/Testing

Distance	Sound Detected?	Average Angle Error (degrees)	Number of Trials
0.25 m - front (0 degrees)	Yes	±6	30
0.75 m - front (0 degrees)	Yes	±7	30
1 m - front (0 degrees)	Yes	±10	30
3 m - front (0 degrees)	Yes	±11	30
0.25 m - side (90 degrees)	Yes	±13	30
0.75 m - side (90 degrees)	Yes	±18	30
1 m - side (90 degrees)	Yes	±8	30
3 m - side (90 degrees)	Yes	±16	30

- Furthest distance robot can detect sound: 10 ft

Microphone Subsystem Validation/Testing

Distance	Sound Detected?	Average Angle Error (degrees)	Number of Trials
0.25 m - back (180 degrees)	Yes	±20	30
0.75 m - back (180 degrees)	Yes	±20	30
1 m - back (180 degrees)	Yes	±25	30
3 m - back (180 degrees)	Yes	±8	30
0.25 m - back (270 degrees)	Yes	±6	30
0.75 m - back (270 degrees)	Yes	±10	30
1 m - back (270 degrees)	Yes	±11	30
3 m - back (270 degrees)	Yes	±16	30

Final Integration of Systems